A

MICRO PROJECT

REPORT

On

"Classroom Automation and Monitoring System"

SUBMITTED TO AN AUTONOMOUS INSTITUTE, AFFILIATED TO SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS

S. Y. B. TECH

in

Electronics & Telecommunication Engineering

SUBMITTED BY

Under the Guidance of

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CERTIFICATE

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This Micro Project report is submitted in partial fulfilment of the requirements of **Second Year Bachelor of Technology (B.Tech.) in Electronics & Telecommunication Engineering** under an Autonomous Institute affiliated to **Savitribai Phule Pune University**, Pune, for the academic year **2025–26** as a Microproject Course.

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DECLARATION BY THE STUDENTS

We declare that the Micro Project entitled **"Classroom Automation and Monitoring System"** submitted by us is a record of the work carried out during the period from **January 2026 to April 2026** under the guidance of **Dr. Manisha Waje**.

We further declare that this work has not been submitted previously, in part or in full, for the award of any degree, diploma, associateship, fellowship, or any other academic recognition in this or any other institution.

We also declare that all sources of information and material used in this project have been properly acknowledged in the report.

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ABSTRACT

Classroom environments in many educational institutions still rely on manual control of electrical appliances and traditional attendance systems, leading to energy wastage and increased workload for teachers. To address these challenges, this project presents a Classroom Automation and Monitoring System based on an embedded IoT platform. The proposed system integrates multiple sensors and modules to automate classroom operations and enhance efficiency.

The system utilizes a PIR motion sensor to detect the presence of students. Based on occupancy, lights and fans are automatically controlled using relay modules, thereby reducing unnecessary power consumption. For attendance management, an ESP32-CAM module is employed to capture images and perform face recognition, enabling automatic and accurate attendance recording. The collected attendance data can be stored and accessed for future reference.

Additionally, a DHT11 sensor is incorporated to monitor classroom temperature and humidity in real time, ensuring a comfortable learning environment. All components are interfaced with an ESP32 microcontroller, which acts as the central processing unit, coordinating sensor data processing and device control. The system can also support wireless communication for data monitoring and storage.

The proposed solution is cost-effective, energy-efficient, and easy to implement, making it suitable for smart classroom applications. Experimental results demonstrate improved energy management, reduced manual effort, and enhanced classroom monitoring. This system contributes to the development of smart educational infrastructure by integrating automation and IoT technologies.

Keywords – Classroom Automation, IoT, ESP32, PIR Sensor, Face Recognition, DHT11, Smart Classroom, Energy Efficiency

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LIST OF ABBREVIATIONS

ABBREVIATION	ILLUSTRATION
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
ESP32-CAM	ESP32 Camera Module
PIR	Passive Infrared Senso
DHT11	Temperature and Humidity Sensor

Chapter 1

Introduction

Introduction

In the modern era, rapid advancements in technology have led to the development of smart systems that improve efficiency, reduce human effort, and optimize resource utilization. One such important application is the **classroom automation and monitoring system**, which integrates Internet of Things (IoT), sensors, and embedded systems to create an intelligent learning environment. These systems aim to enhance classroom management, improve student comfort, and reduce energy consumption.[2]

Traditional classroom environments depend largely on manual operations such as controlling lights and fans, maintaining attendance records, and monitoring environmental conditions. These conventional methods are time-consuming, inefficient, and often lead to errors and unnecessary energy wastage. With the introduction of IoT and automation technologies, classrooms can now be transformed into smart environments where devices operate automatically based on real-time data. [2,4,6]

Several research studies have explored different aspects of classroom automation. One study proposed an IoT-based smart classroom system integrated with facial recognition and automated lighting. This system enables automatic attendance marking and efficient energy management, reducing manual effort and power consumption. However, challenges such as scalability and implementation in large institutions were identified. [1,3]

Another research work focused on monitoring classroom environmental conditions using sensors to measure temperature, humidity, and air quality. Maintaining optimal environmental conditions helps improve student comfort and enhances learning performance. Despite its benefits, the system offers limited automation and lacks intelligent decision-making features.

In addition, a smart attendance and monitoring system was developed using technologies such as ARM processors, RFID, GSM, and Bluetooth. This system provides efficient classroom management by automating attendance and communication processes. However, the complexity of the system and the requirement of cloud integration make it less suitable for simple and low-cost implementations.[2]

A different approach introduced an IoT-based seat allocation system using Arduino and RFID technologies. This system improves accuracy and reduces time required for managing seating arrangements in classrooms. However, it lacks advanced features such as data analytics and integration with other classroom automation components. [4,6]

Energy efficiency has also been addressed through the development of smart lighting systems that automatically control lights based on occupancy and light intensity. Such systems significantly reduce energy consumption by ensuring that electrical devices operate only when required. However, these systems are often limited to specific functionalities and are not integrated with other classroom automation features.[5]

Another study proposed a Raspberry Pi and smartphone-based classroom automation system for device control and monitoring. This system enhances security and automation but may increase overall system cost, making it less feasible for large-scale deployment. Therefore, there is a need to design cost-effective solutions that can be implemented widely.

From the review of these existing techniques, although various solutions have been developed, most systems focus on individual functionalities such as attendance, lighting, or environmental monitoring. There is a lack of a fully integrated system that combines all these features into a single platform. Additionally, issues such as high cost, system complexity, limited scalability, and lack of advanced analytics remain key challenges.

To overcome these limitations, the proposed classroom automation system aims to develop a **comprehensive, cost-effective, and scalable solution** by integrating multiple functionalities such as automated attendance, environmental monitoring, smart lighting. By using modern technologies like IoT and ESP32, the system can provide real-time monitoring and control, improve energy efficiency, and enhance the overall classroom experience.

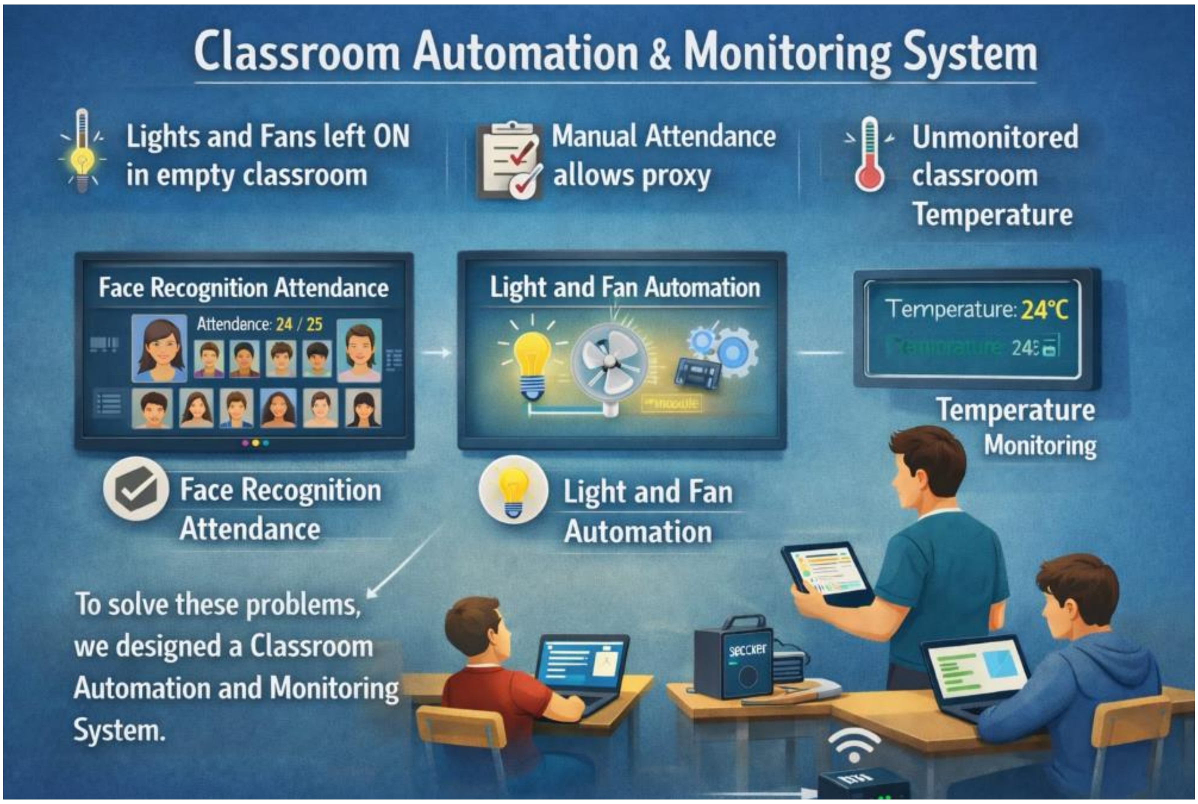

Fig 1.1 Classroom Automation and Monitoring System.

1.1 Background and Motivation

With the advancement of Internet of Things (IoT) and smart technologies, traditional systems are gradually being replaced by automated and intelligent solutions. In the context of education, classroom environments still rely on manual operations such as controlling electrical devices, taking attendance, and monitoring environmental conditions. These conventional methods are time-consuming, inefficient, and often lead to human errors and energy wastage.

To address these issues, various smart classroom systems have been developed. An IoT-based classroom system with automated attendance and smart lighting improves energy efficiency and reduces manual effort. Similarly, environmental monitoring systems using sensors help maintain suitable temperature, humidity, and air quality, thereby enhancing student comfort and performance. However, these systems mainly focus on monitoring and lack full automation.

Automated attendance systems using RFID, GSM, and Bluetooth technologies have been introduced to improve classroom management. While these systems are efficient, they are often complex and require additional infrastructure. Seat allocation systems based on IoT and RFID technologies have also been developed to improve accuracy and save time, but they lack integration with other classroom features.

Energy-saving solutions such as smart lighting systems automatically control lights based on occupancy and intensity, significantly reducing power consumption. In addition, Raspberry Pi-based classroom automation systems provide better control and security through smartphone-based applications. However, these systems can be costly and are not always suitable for large-scale implementation.

From the study of these existing systems, it is observed that most solutions focus on individual functionalities rather than providing a complete and integrated system. Challenges such as high cost, system complexity, limited scalability, and lack of comprehensive automation still remain.

Therefore, the motivation behind this project is to develop a **cost-effective, scalable, and integrated classroom automation system** that combines multiple features such as automated attendance, environmental monitoring, and smart device control. This system aims to improve efficiency, reduce human effort, and create a smart and energy-efficient classroom environment.

1.2 Need

In today's educational environment, the need for smart and automated systems is increasing due to the limitations of traditional classroom methods. Conventional classrooms depend on manual operations for controlling electrical appliances, managing attendance, and monitoring environmental conditions. These processes are not only time-consuming but also lead to inefficiency, human errors, and unnecessary energy consumption.

One of the major needs for a classroom automation system is **efficient attendance management**. Manual attendance marking is prone to errors and consumes valuable lecture time. Although automated attendance systems using technologies such as RFID and facial recognition have been developed, they are often complex and not fully integrated with other classroom functions. Hence, there is a need for a simplified and integrated solution.
Another important requirement is **energy management**. In many classrooms, lights and fans are left ON even when not required, leading to significant energy wastage. Smart lighting systems based on occupancy and intensity have shown improvements in reducing energy consumption. However, these systems are usually limited to lighting control only. Therefore, a more comprehensive system is needed to manage all electrical devices efficiently.
Maintaining a proper **classroom environment** is also essential for effective learning. Factors such as temperature, humidity, and air quality directly affect student comfort and concentration. Existing systems use sensors to monitor these parameters and improve classroom conditions, but they lack automation and intelligent control mechanisms. This creates a need for systems that not only monitor but also automatically adjust environmental conditions.
In addition, there is a growing need for **centralized control and monitoring** of classroom devices.

1.3 Significance and Future Applications

One of the key significances of the system is **automation of routine tasks** such as attendance marking and device control. Existing systems have demonstrated that automated attendance using technologies like facial recognition and RFID can save time and improve accuracy. This allows teachers to focus more on teaching rather than administrative tasks.

Another important aspect is **energy efficiency**. Smart lighting and device control systems reduce unnecessary power consumption by operating appliances only when required. This not only lowers electricity costs but also contributes to environmental sustainability, which is an essential requirement in modern smart infrastructures.

The system also enhances the **learning environment** by monitoring parameters such as temperature, humidity, and air quality. Maintaining a comfortable classroom environment improves student concentration, participation, and overall academic performance. In addition, automated monitoring ensures that conditions are maintained without constant manual supervision.

Furthermore, the proposed system provides **centralized monitoring and control**, which improves classroom management and security. Systems based on Raspberry Pi and mobile applications have shown that remote access and control can make classroom operations more convenient and efficient. This feature is especially useful for managing multiple classrooms within an institution.

Another significant advantage is the **integration of multiple functionalities** into a single system. Unlike existing solutions that focus on individual features such as attendance or seating arrangements, the proposed system combines various components to create a unified and efficient solution. This reduces system complexity and improves overall performance.

Future Applications

The scope of classroom automation systems extends beyond basic applications and can be expanded in the future with advanced technologies. One potential application is the integration of **Artificial Intelligence (AI)** for smart decision-making, such as predicting student attendance patterns or automatically adjusting environmental conditions.

The system can also be enhanced with **cloud computing and data analytics**, allowing storage and analysis of classroom data for better decision-making and performance evaluation. This can help institutions track student progress and improve teaching strategies.

Another future application is the development of **smart campuses**, where multiple classrooms and facilities are connected through a centralized IoT network. This will enable efficient management of resources across the entire institution.

Additionally, the system can be integrated with **mobile applications** to provide real-time updates to teachers and students regarding attendance, classroom conditions, and schedules. Advanced features such as voice control and biometric authentication can further improve usability and security.

In conclusion, the proposed classroom automation system is highly significant in modern education as it improves efficiency, reduces energy consumption, and enhances the overall learning experience. With future advancements, it has the potential to evolve into a fully intelligent and connected educational ecosystem.

Chapter 2

Literature Review

Literature Review

Table 2.1: Literature Review

Sr. No.	Project Title	Author	Published Year	Outcome	Limitation/future scope
1.	IT-Based Smart Classroom with Facial Recognition	Mohammed Adu Rahman	(2021) Used IoT system with facial recognition, sensors and Al algorithms for automated classroom and attendance monitoring	Improves attendance accuracy and reduces manual effort	High cost and privacy concerns →Can improve by using low-cost Al models and secure cloud storage.
2.	Smart Classroom Environment Monitoring System	Mustafa & Duraklar	(2022) Implemented temperature, humidity and air quality sensors	Maintains comfortable classroom conditions automatically	Limited automation → Can improve by adding AI-based decision control
3.	Smart Attendance and Monitoring System	Vishnu Priya Reddy	(2022) Used ARM, RFID, GSM and Bluetooth for attendance system	Provides efficient and automated attendance monitoring	Complex hardware setup → Can improve using cloud integration and scalable models
4.	IoT-Based Classroom Seat Allocation System	O.E. Ojep et al.	(2021) Developed Arduino-based fingerprint sensor system	Increases seating accuracy and saves lecturer's time	Hardware limitations → Can improve by enhancing analytics and real-time updates
5.	Smart Lighting System	Baoshi Sun et al.	(2021) Developed IoT-based smart lighting using sensors	Reduces energy consumption effectively	Hardware limitations → Can improve by enhancing analytics and real-time updates
6.	IoT-Based Classroom Seat Allocation System	Dr. Eze Chimdy et al.	(2022) Designed Raspberry Pi and smartphone-based system	Enhances classroom security and automation	High implementation cost → Can improve by designing low-cost scalable model

2.1 Evolution of Existing techniques

The concept of classroom management has evolved significantly over time, moving from completely manual systems to advanced automated and intelligent solutions. Earlier, traditional classrooms relied entirely on human effort, where teachers or staff manually controlled electrical appliances such as lights and fans and maintained attendance records. This approach was time-consuming, prone to human error, and often resulted in unnecessary energy consumption due to lack of proper monitoring

.

With the advancement of electronics and embedded systems, semi-automated solutions were introduced using microcontrollers and basic sensors. These systems utilized motion detectors and infrared (IR) sensors to control lighting and fan operations based on occupancy. Such techniques improved energy efficiency and reduced manual intervention to some extent, as discussed in. However, these systems were limited in functionality and lacked real-time monitoring capabilities.

The introduction of Internet of Things (IoT) technology marked a major shift in classroom automation. IoT-enabled systems allowed devices to communicate over the internet, enabling remote monitoring and control of classroom environments. Researchers developed smart systems that could monitor parameters such as temperature, humidity, and air quality, improving comfort and learning conditions. These systems also allowed data collection and analysis, making them more efficient and intelligent compared to earlier methods.

Further advancements led to the integration of camera modules and image processing techniques for automated attendance systems. Facial recognition-based systems were introduced to eliminate manual attendance marking and improve accuracy. These systems enhanced security and reduced proxy attendance but also introduced challenges related to privacy and computational requirements.

Recent developments focus on combining IoT with Artificial Intelligence (AI) and cloud computing to create fully smart classroom environments. These systems not only automate appliances but also provide real-time analytics, remote access through mobile applications, and intelligent decision-making capabilities. Despite these improvements, existing techniques still face issues such as scalability, cost, and dependency on network connectivity.

Overall, the evolution of classroom automation systems reflects a transition from manual operations to intelligent, connected, and data-driven solutions, aiming to improve efficiency, reduce energy consumption, and enhance the overall learning experience

2.2 Summary and Research Gap

From the review of existing classroom automation systems, it is observed that significant progress has been made in improving classroom efficiency through the use of sensors, IoT, and smart monitoring techniques. Earlier systems focused mainly on basic automation such as controlling lights and fans using motion or IR sensors, which helped in reducing energy consumption but lacked intelligence and adaptability.

With the introduction of IoT-based systems, researchers were able to implement real-time monitoring and remote control of classroom environments. These systems improved comfort levels by monitoring temperature, humidity, and air quality. Additionally, camera-based systems and facial recognition techniques enhanced attendance management by making it automatic and more accurate.

However, despite these advancements, several research gaps still exist. Most existing systems provide limited automation, focusing only on specific tasks like lighting or attendance rather than offering a fully integrated solution. Many systems also depend heavily on stable internet connectivity, which can affect performance in real-time applications.

Another major gap is the lack of advanced intelligence in decision-making. While some systems use basic automation, they do not incorporate Artificial Intelligence for predictive analysis or smart control. Privacy and security concerns related to camera-based monitoring systems are also not fully addressed in many studies.

Furthermore, issues such as scalability, cost, and system complexity make it difficult to implement these solutions in all educational institutions. Some systems require high initial setup and technical expertise, limiting their practical usability.

Therefore, there is a need to develop a cost-effective, reliable, and fully integrated classroom automation system that combines IoT, smart monitoring, and improved control mechanisms while addressing limitations such as privacy, scalability, and dependency on network connectivity.

2.3 Problem Statement

Modern classrooms still rely heavily on manual operation of electrical appliances and attendance management systems. This leads to inefficient energy usage, increased workload for teachers, and lack of proper monitoring. In many cases, lights and fans remain switched ON even when the classroom is empty, resulting in unnecessary power wastage.
Additionally, traditional attendance methods are prone to errors and proxy entries, reducing reliability. Existing automation systems are either partially implemented, expensive, or dependent on continuous internet connectivity. They also lack intelligent decision-making and raise privacy concerns when camera-based monitoring is used.
Therefore, there is a need to develop a smart, efficient, and cost-effective classroom automation and monitoring system that overcomes these limitations while improving overall classroom management.

Objectives

The main objectives of the proposed system are:

1.To automate the control of classroom appliances like lights and fans based on occupancy
2. To reduce energy consumption and improve power efficiency
3. To implement an accurate and reliable attendance management system
4. To provide real-time monitoring of classroom activities

Chapter 3

System Specifications

System Specification

3.1 Components used

The proposed classroom automation and monitoring system uses the following hardware components:

1. **ESP32 (38-Pin NodeMCU):**
   It acts as the main controller of the system. It processes input from sensors and controls output devices. It also provides built-in Wi-Fi and Bluetooth for IoT connectivity.

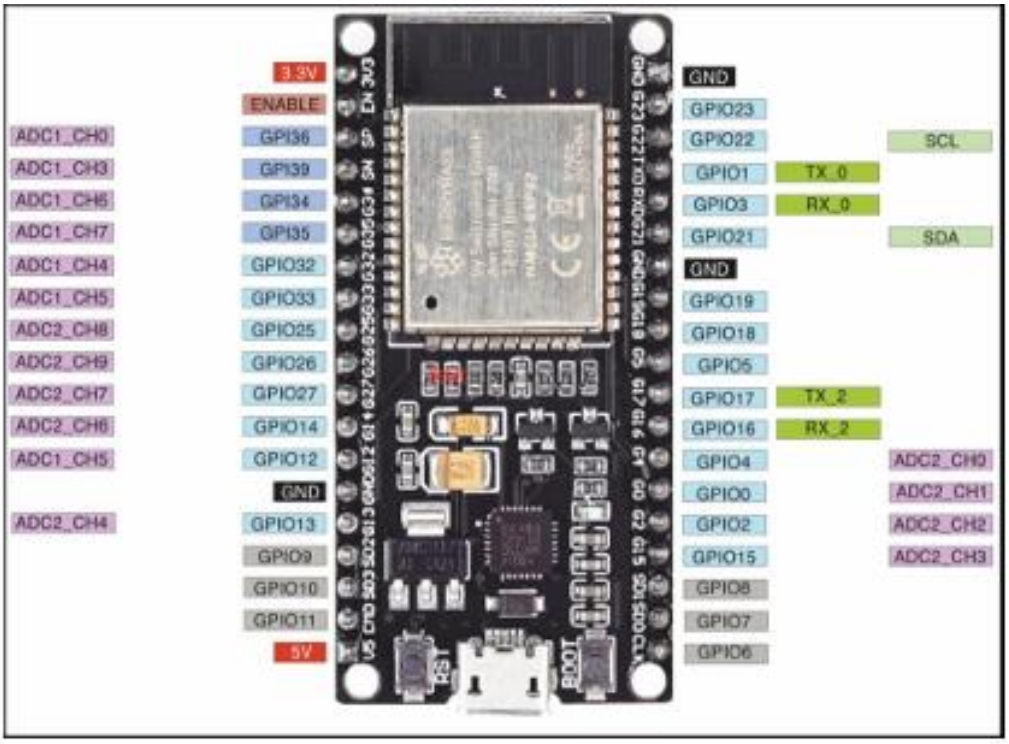

Fig 3.1 Pin configuration of ESP32 (38-Pin NodeMCU)

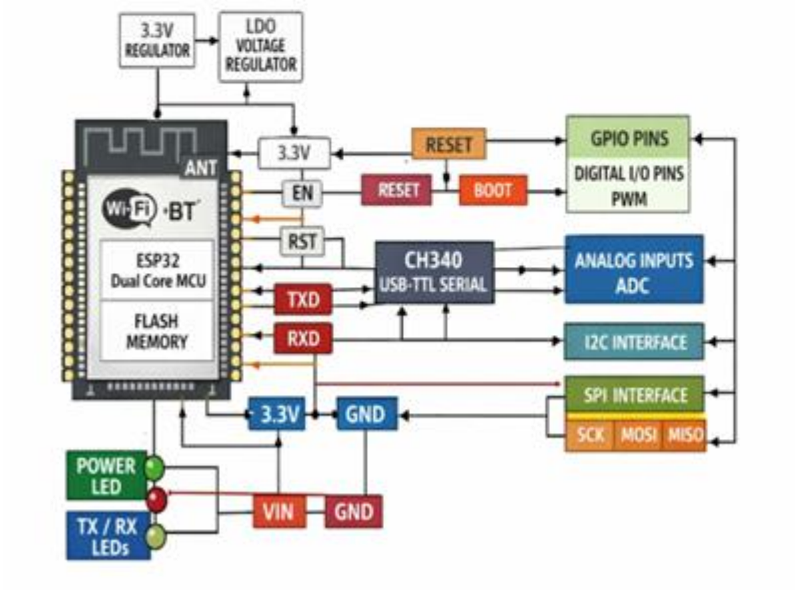

Fig 3.2 Internal circuit diagram of ESP32 (38-Pin NodeMCU)

2. **ESP32-CAM Module (OV3660 Camera):**
   This module is used for capturing images and video for real-time classroom monitoring. It enables remote surveillance and can be used for attendance purposes.

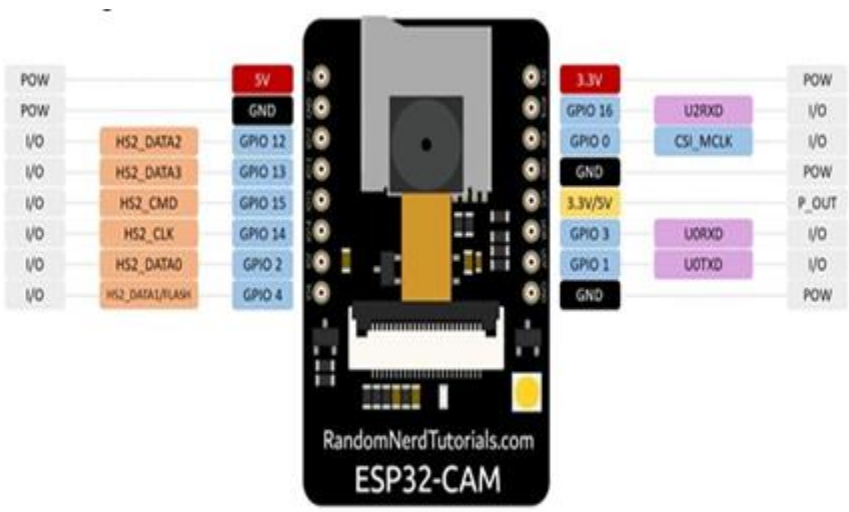

Fig 3.3 Pin configuration of ESP32-CAM Module (OV3660 Camera)

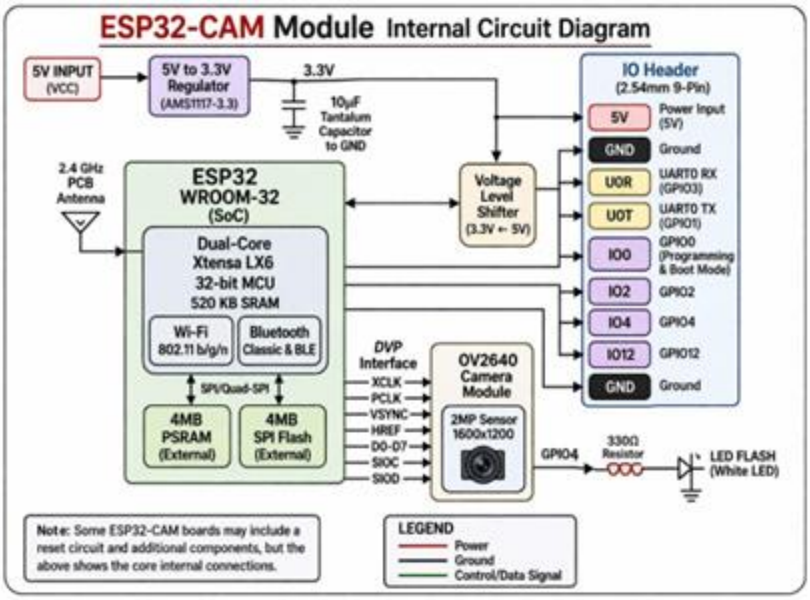

Fig 3.4 Internal circuit diagram of ESP32-CAM Module (OV3660 Camera)

3. **PIR Sensor (Passive Infrared Sensor):**
   Detects human presence based on body heat and motion. It helps in automatically controlling lights and fans.

Fig 3.5 Pin configuration of PIR Sensor

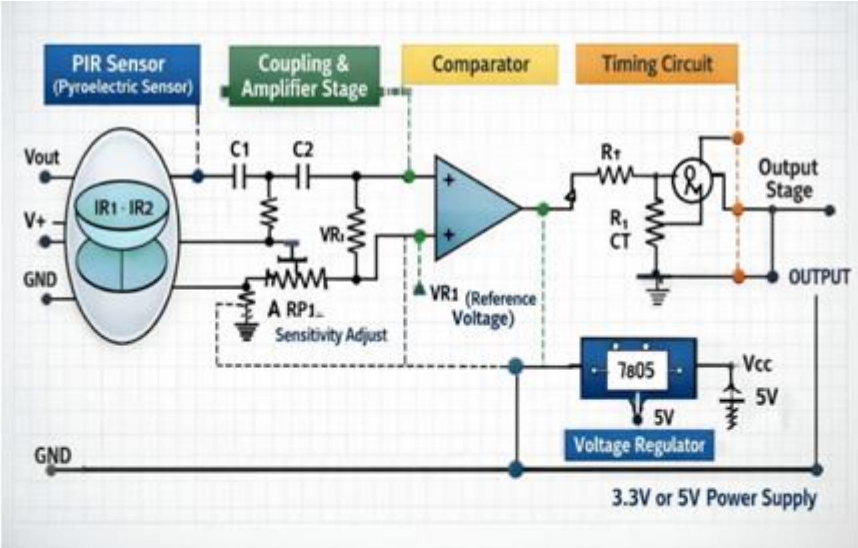

Fig 3.6 Internal circuit diagram of PIR Sensor

4. DHT 11 Sensor:

A 4-pin, single-row DHT11 sensor features VCC (3.3-5V), Data, No Connect (NC), and Ground pins in that order from left to right, typically requiring a pull-up resistor between the Data and VCC pins for communication.

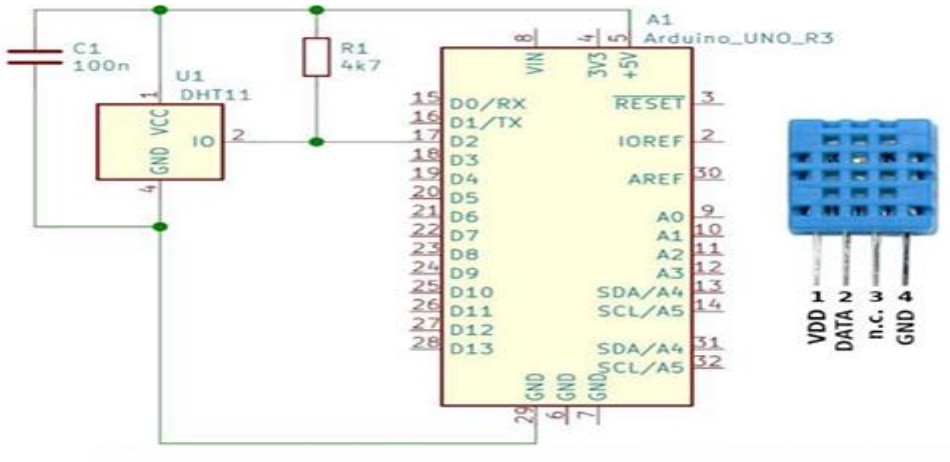

Fig 3.7 Pin configuration of DHT 11 Sensor

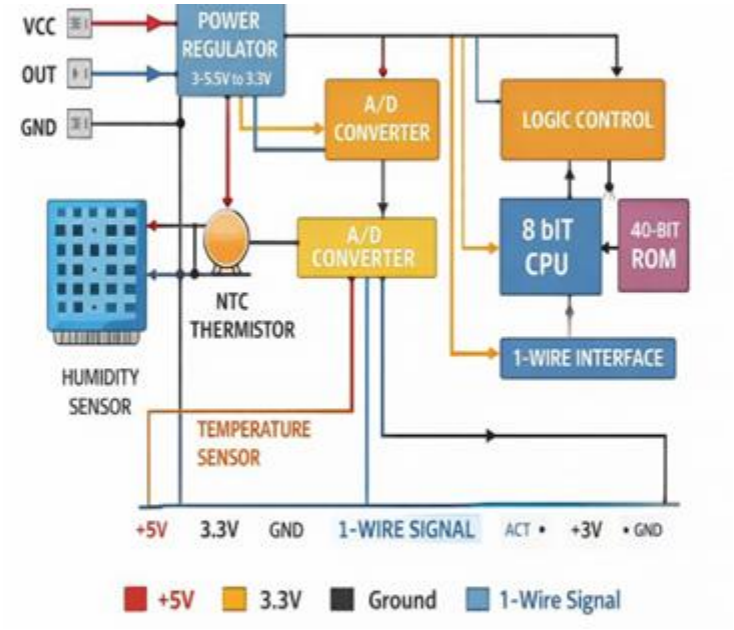

Fig 3.8 Internal circuit diagram of DHT 11 Sensor

5. Relay Module:
Acts as an electronic switch to control high-voltage devices like lights and fans based on signals from the ESP32.

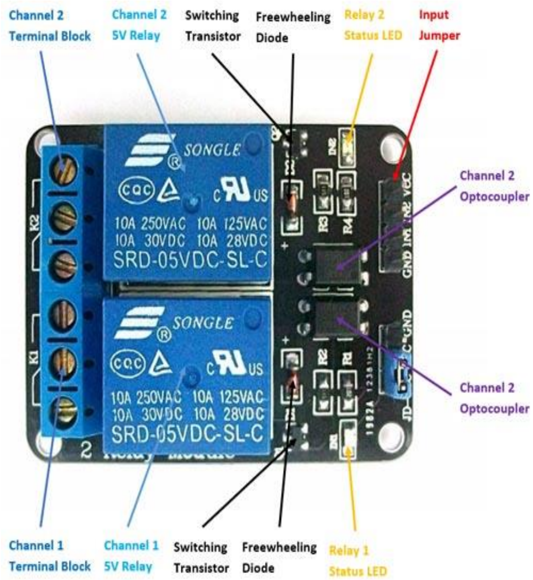

Fig 3.9 Pin configuration of Relay Module

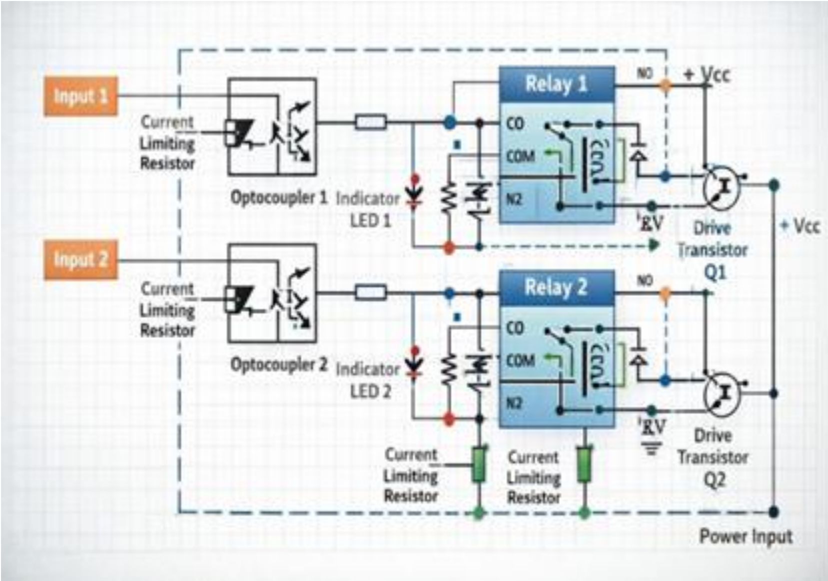

Fig 3.10 Internal circuit diagram of Relay Module

5. **Power Supply Unit:**
   Provides required power to all components in the system.

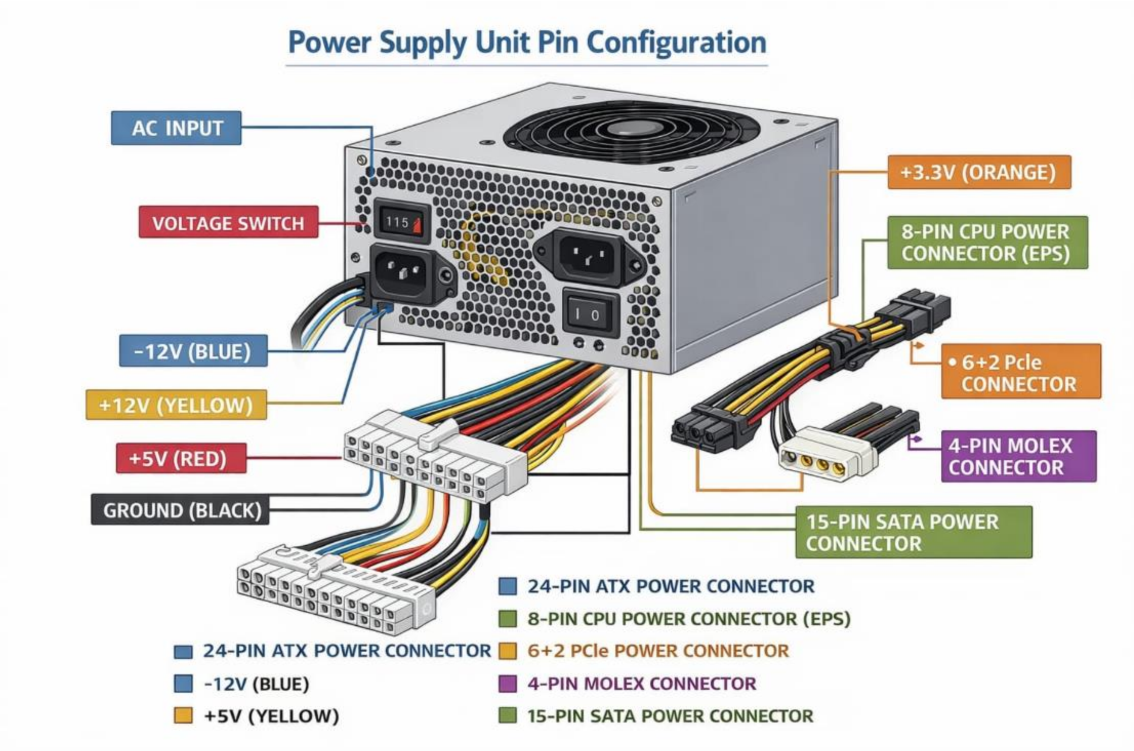

Fig 3.11 Pin configuration of Power Supply Unit

6. **Connecting Wires and Breadboard:**
   Used for circuit connections and prototyping of the system

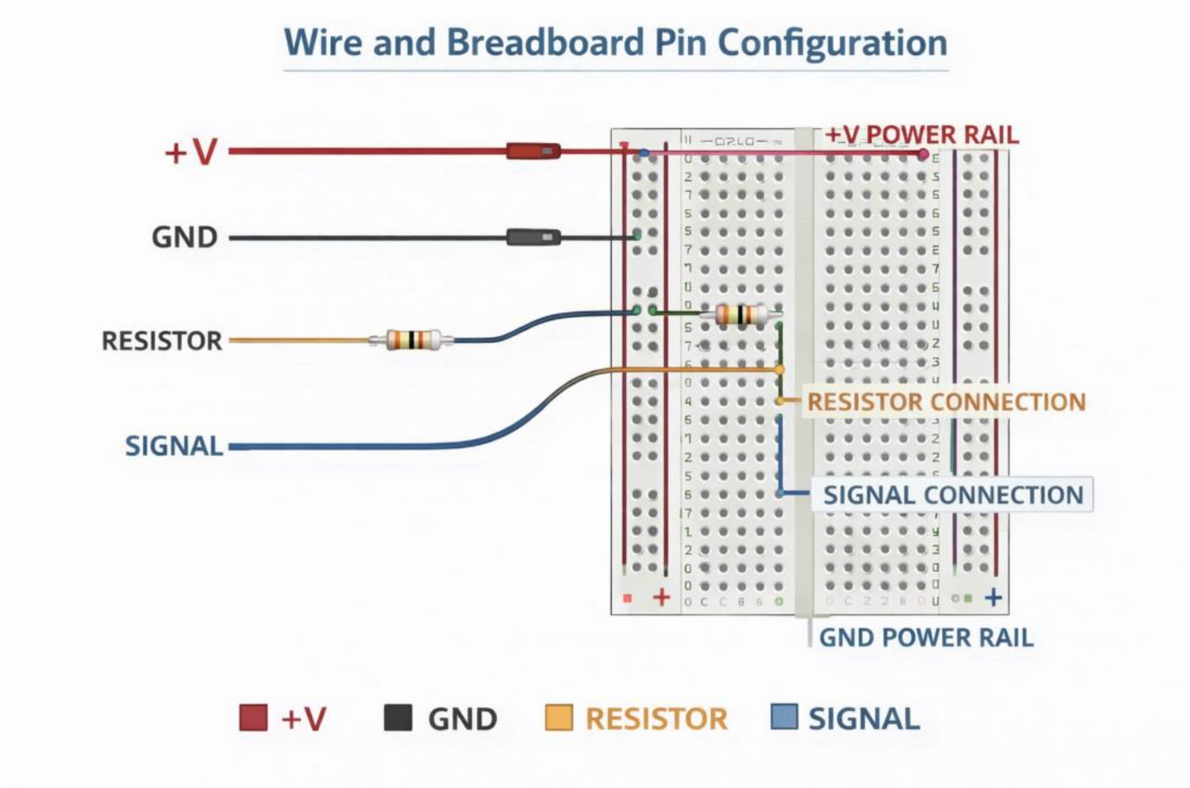

Fig 3.12 Pin configuration of Connecting Wires and Breadboard

3.2 Software used

The system is developed using the following software tools:

1. **Arduino IDE:**
   Used for writing, compiling, and uploading code to the ESP32 and ESP32-CAM modules.
2. **Embedded C / Arduino Programming Language:**
   Used to program the logic for sensor input, device control, and communication
3. **Wi-Fi / IoT Platform**
   Enables communication between the system and user interface for remote monitoring and control.
4. **Web Interface / Mobile Application (if used):**
   Provides a user-friendly interface to monitor classroom status and control devices remotely.

Chapter 4

Proposed Work

Proposed Work

The proposed system aims to develop a **classroom automation and monitoring system** using IoT technology to improve energy efficiency, reduce manual effort, and enhance classroom management.

The system is designed using **ESP32 (NodeMCU)** as the main controller and **ESP32-CAM module** for real-time monitoring. Sensors such as **PIR and IR sensors** are used to detect the presence of students inside the classroom. Based on this detection, the system automatically controls electrical appliances like lights and fans.

When a person enters the classroom, the PIR sensor detects motion and sends a signal to the ESP32. The controller then activates the relay module, which turns ON the lights and fans. If no motion is detected for a certain period, the system automatically switches OFF the appliances, thereby saving energy.

The ESP32-CAM module is used to capture images or provide live streaming of classroom activity. This enables remote monitoring through a web interface or mobile application using Wi-Fi connectivity. The system ensures real-time observation without the need for physical presence.

Additionally, the system can be extended for **attendance monitoring** by integrating image processing techniques. This reduces manual attendance errors and improves efficiency.

The proposed system focuses on being **cost-effective, reliable, and easy to implement**, making it suitable for educational institutions. It also addresses key limitations of existing systems by providing integrated functionality, reduced power consumption, and minimal human intervention.

Key Features of Proposed System

1. Automatic control of lights and fans
2. Real-time classroom monitoring using ESP32-CAM
3. Energy-efficient operation
4. Reduced human effort
5. Scalable and cost-effective design

4.1 Block Diagram

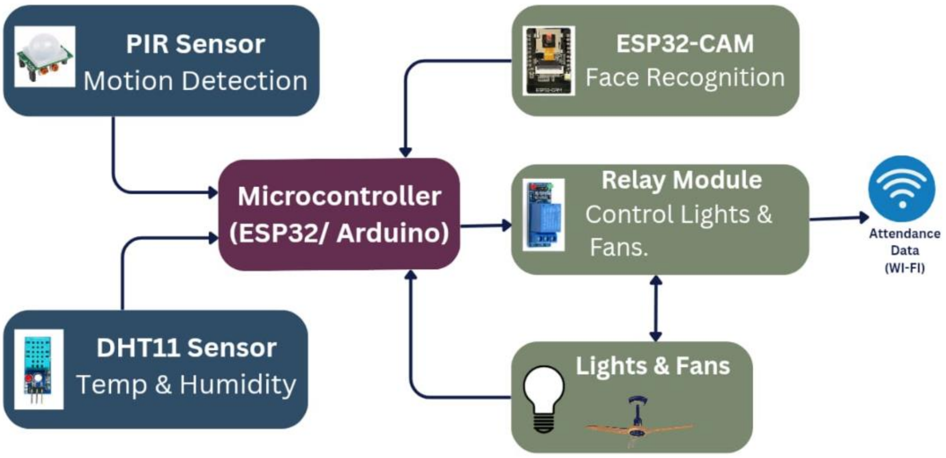

Fig 4.1 Block diagram

4.2 Flow Chart

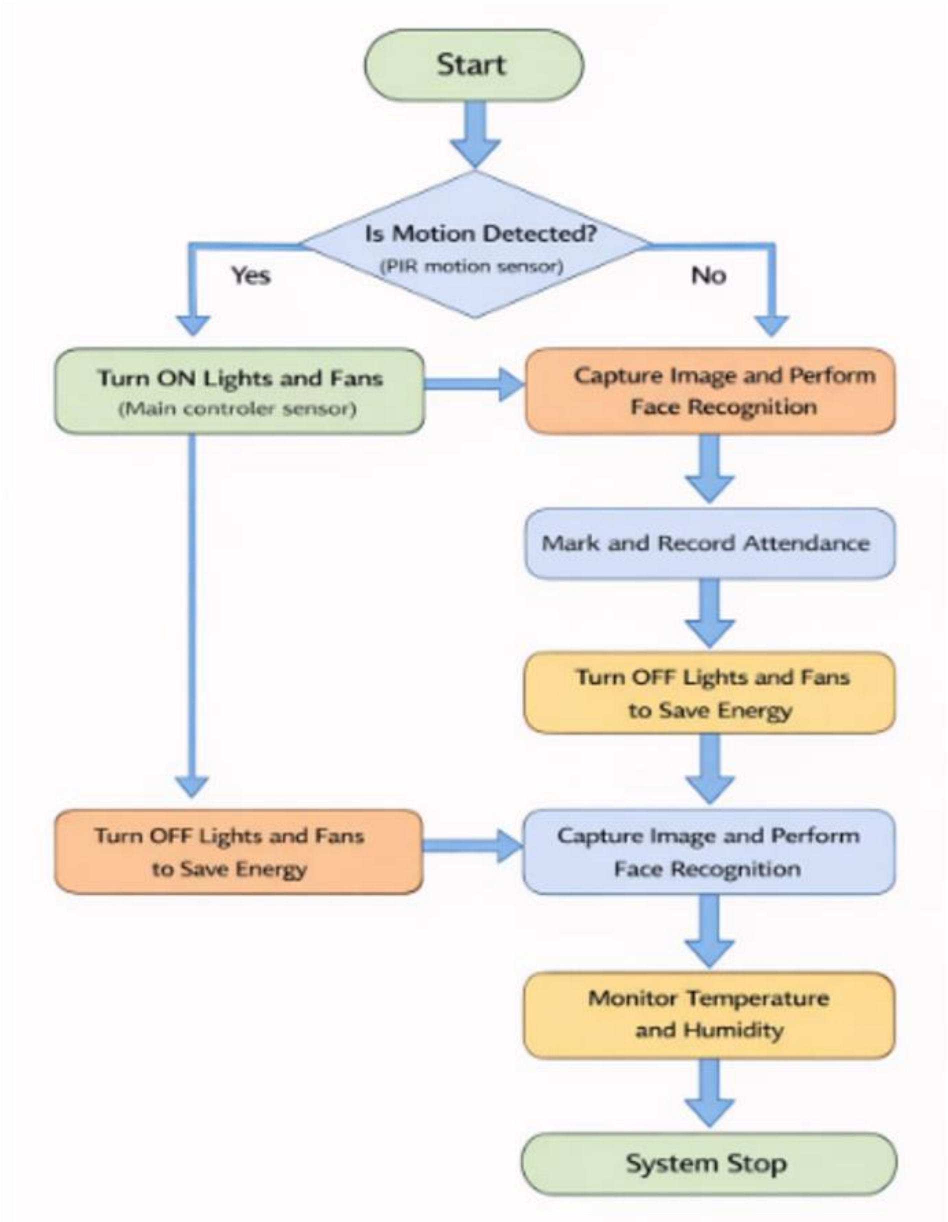

Fig 4.2 Flow Chart

4.3 Circuit Diagram

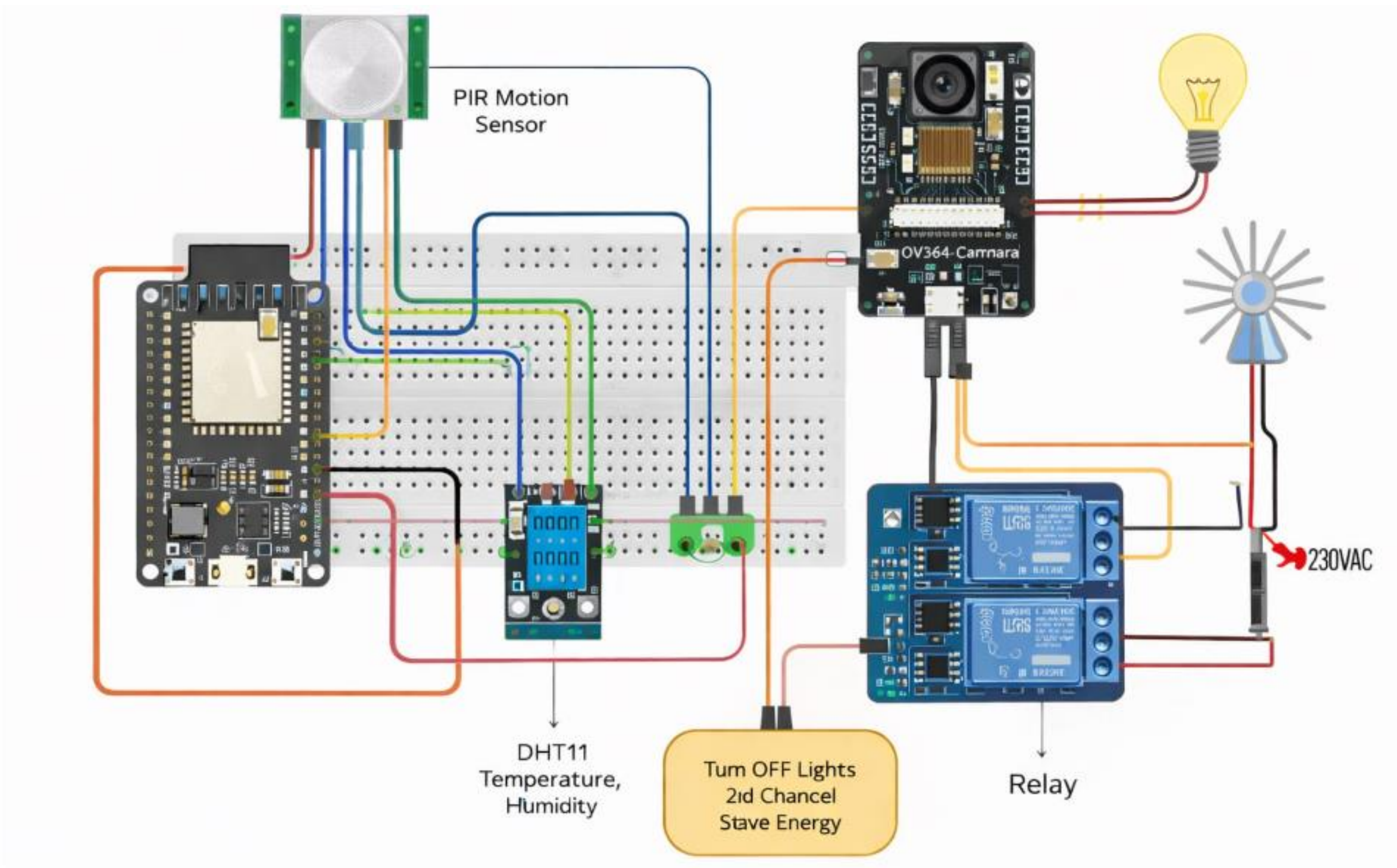

Fig 4.3 Circuit Diagram

Chapter 5

Result and Analysis

Results and analysis

The smart classroom automation and monitoring system was successfully designed and implemented. The system demonstrated effective automatic control of electrical appliances such as lights and fans based on occupancy detection.

The attendance system worked accurately, reducing manual errors and preventing proxy attendance. Real-time monitoring using IoT modules enabled users to observe classroom conditions efficiently. The system operated with minimal human intervention and showed improved energy efficiency by turning OFF appliances when the classroom was empty.

Overall, the system performed reliably under test conditions and met the intended design requirements.

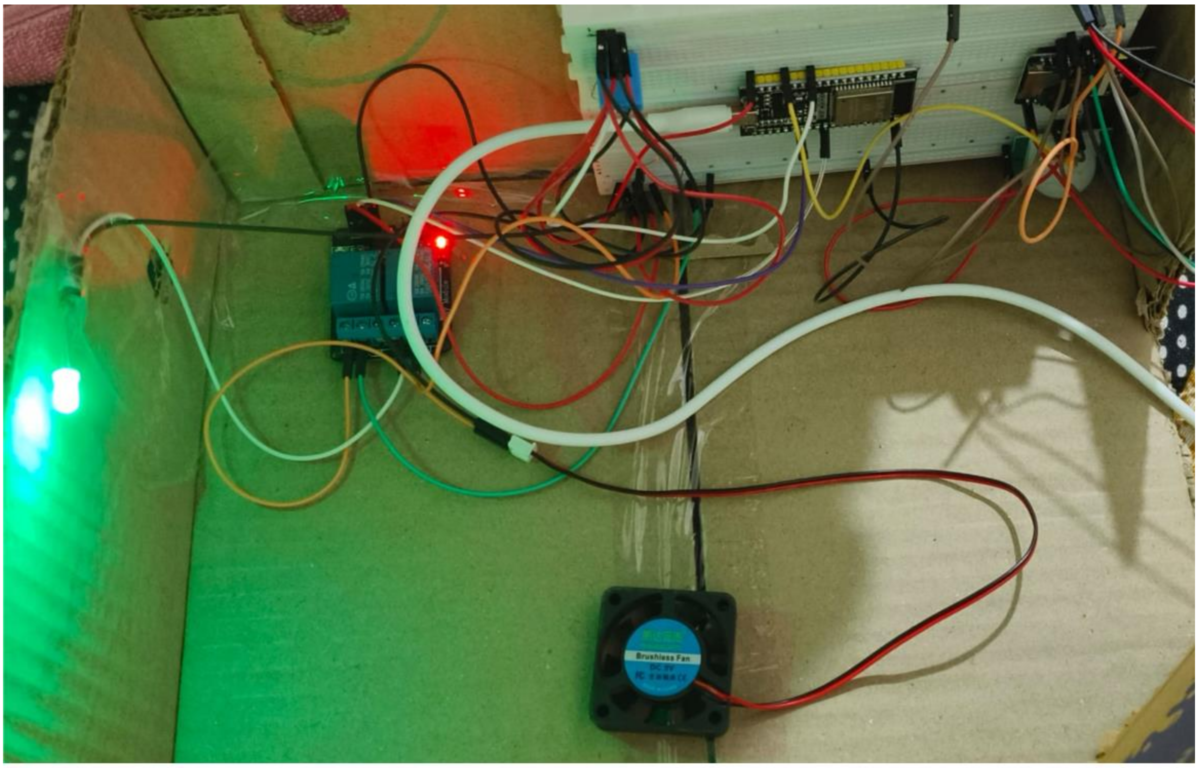

Fig 5.1 Result

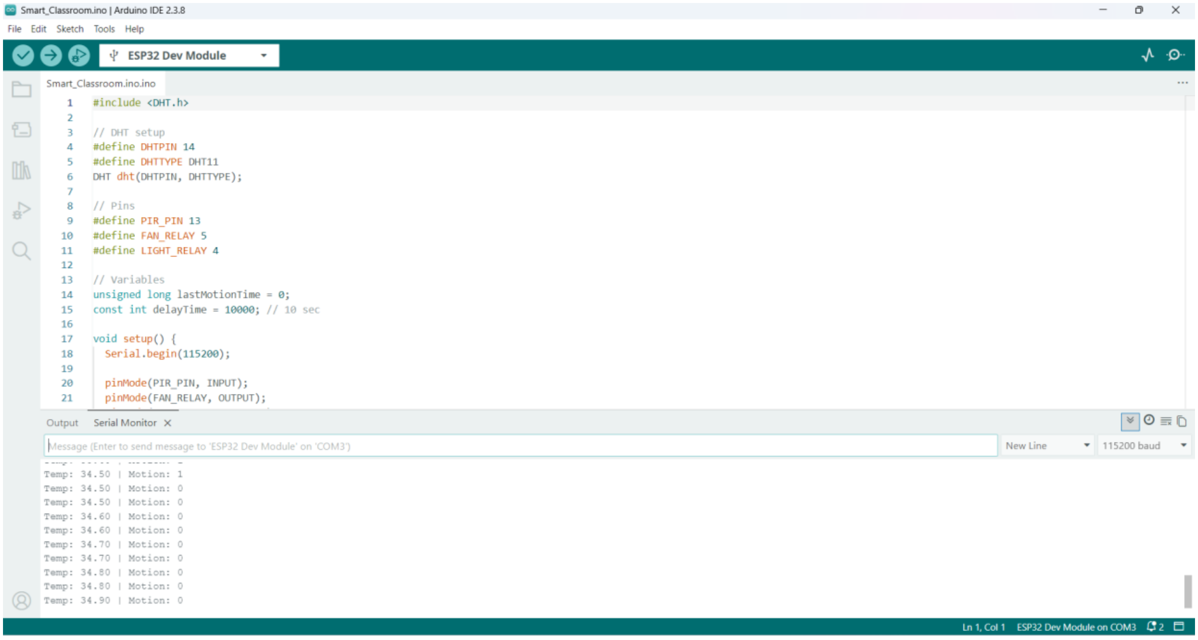

Fig 5.2 Arduino Code

The performance of the system was analysed based on efficiency, accuracy, cost, and usability.

1. Energy Efficiency:
   The system significantly reduced power wastage by automatically controlling appliances based on occupancy.
2. Accuracy:
   Attendance tracking and sensor-based detection provided reliable and consistent results compared to manual methods.
3. Cost Effectiveness:
   The use of affordable components like microcontrollers and sensors made the system economical.
4. Reliability:
   The system worked efficiently even with limited internet dependency, improving stability.
5. User-Friendliness:
   The interface and operation were simple, making it easy for users to manage and monitor the system.
6. Limitations Observed:
   Minor delays in sensor response and dependency on hardware accuracy were noted. Environmental factors like lighting conditions could affect sensor performance.

Chapter 6

Conclusion

Conclusion

This project, Classroom Automation and Monitoring System, was successfully designed and implemented using ESP32, PIR sensor, DHT11 sensor, relay module, and camera module. The system is able to automatically control classroom appliances like lights and fans based on real-time conditions.

When motion is detected by the PIR sensor, the system turns ON the lights, which helps in saving electricity when the classroom is empty. The DHT11 sensor monitors the temperature, and based on that, the fan is controlled automatically. In addition to automation, the ESP32-CAM provides live monitoring through Wi-Fi, which increases the security and supervision of the classroom.

While working on this project, we understood practical concepts like sensor interfacing, GPIO control, and wireless communication. We also faced some challenges like connection issues and power supply stability, but we solved them through testing and troubleshooting.
Overall, this project is a useful and cost-effective solution for smart classroom management, and it can be further improved by adding features like mobile app control or face recognition for attendance.

Chapter 7

Future Scope

Future Scope

1. Integration of AI for intelligent control of classroom devices
2. Face recognition system for automatic attendance
3. Mobile app-based monitoring and control
4. Voice control using assistants like Google Assistant
5. Cloud-based data storage and real-time monitoring

References

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[4] O.E. Ojep et al., "IoT-Based Classroom Automation System," *IEEE* 2021.

[5] Baoshi Sun et al., "Smart Lighting System using IoT," *IEEE* 2021.

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APPENDIX

A. Datasheets

1. ESP32 Microcontroller

Table 7.1 ESP32 Microcontroller Specification

Parameter	Specification
Operating Voltage	3.3V
Processor	Dual-core Xtensa LX6
Clock Frequency	Up to 240 MHz
Digital I/O Pins	30+
Analog Pins	12-bit ADC
Communication	WiFi, Bluetooth, UART, I2C, SPI

2. ESP32-CAM Module

Table 7.2 ESP32 -CAM Module Specification

Parameter	Specification
Operating Voltage	5V
Camera	OV2640 (2MP)
Image Output	JPEG
Connectivity	WiFi, Bluetooth
Storage	MicroSD card supported
Application	Face recognition & image capture

3.PIR Sensor (Motion Sensor)

Table 7.3 PIR Sensor Specification

Parameter	Specification
Operating Voltage	3.3V – 5V
Detection Range	Up to 7 meters
Output Type	Digital (HIGH/LOW)
Delay Time	Adjustable
Sensitivity	Adjustable
Application	Human motion detection

5. DHT11 Temperature Sensor

Table 7.4 DHT11 Temperature Sensor Specification

Parameter	Specification
Operating Voltage	3.3V – 5V
Temperature Range	0°C to 50°C
Humidity Range	20% to 90%
Accuracy	±2°C, ±5% RH
Output	Digital
Application	Temperature & humidity monitoring

6. Relay Module

Table 7.5 Relay Module Specification

Parameter	Specification
Operating Voltage	5V
Channels	1 / 2 Channel
Trigger Type	LOW/HIGH
Load Capacity	250V AC / 10A
Isolation	Optocoupler
Application	Control lights & fans

7. Power Supply

Table 7.6 Power Supply Specification

Parameter	Specification
Input Voltage	230V AC
Output Voltage	5V DC
Type	Adapter / SMPS
Application	Powering system

8. **Connecting Wires & Breadboard**

Table 7.7 Connecting Wires & Breadboard Specification

Parameter	Specification
Type	Jumper wires
Usage	Circuit connections
Breadboard Type	Solderless
Application	Prototyping

B. Pin Connections Table

Table 7.8 ESP32 Microcontroller Pin Connection

Component	Pin (Component)	ESP32 Pin
PIR Sensor	VCC	5V
PIR Sensor	GND	GND
PIR Sensor	OUT	GPIO 13
DHT11 Sensor	VCC	3.3V
DHT11 Sensor	GND	GND
DHT11 Sensor	DATA	GPIO 14
Relay Module (Fan)	VCC	5V
Relay Module (Fan)	GND	GND
Relay Module (Fan)	IN	GPIO 12
Relay Module (Light)	VCC	5V
Relay Module (Light)	GND	GND
Relay Module (Light)	IN	GPIO 27
LED	Anode (+)	GPIO 26
LED	Cathode (-)	GND (via resistor)

C. Arduino Code

```
#include <DHT.h>
// Pins
#define PIR_PIN    13
#define LIGHT_PIN  4
#define FAN_PIN    5
#define DHT_PIN    14
#define DHTTYPE DHT11
DHT dht(DHT_PIN, DHTTYPE);
// Variables for motion
unsigned long lastMotionTime = 0;
bool lightState = false;
void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(LIGHT_PIN, OUTPUT);
  pinMode(FAN_PIN, OUTPUT);
  // Relay OFF (Active LOW)
  digitalWrite(LIGHT_PIN, HIGH);
  digitalWrite(FAN_PIN, HIGH);
  Serial.begin(115200);
  dht.begin();
}

void loop() {
  unsigned long currentTime = millis();
  // ---------------- LIGHT (PIR) ----------------
  int motion = digitalRead(PIR_PIN);
  if (motion == HIGH) {
    lastMotionTime = currentTime;
    if (!lightState) {
      digitalWrite(LIGHT_PIN, LOW); // ON
      lightState = true;
```

```
      Serial.println("Motion detected -> Light ON");
    } else {
      Serial.println("Continuous motion detected -> Light remains ON");
    }
  }
  // No motion for 5 sec
  if (lightState && (currentTime - lastMotionTime >= 5000)) {
    digitalWrite(LIGHT_PIN, HIGH); // OFF
    lightState = false;
    Serial.println("No motion for 5 seconds -> Light OFF");
  }
  // ---------------- FAN (DHT11) ----------------
  float temp = dht.readTemperature();
  if (!isnan(temp)) {
    Serial.print("Temperature: ");
    Serial.println(temp);
    if (temp >= 32) {
      digitalWrite(FAN_PIN, LOW); // ON
      Serial.println("Temperature above 32°C -> Fan ON");
    } else {
      digitalWrite(FAN_PIN, HIGH); // OFF
      Serial.println("Temperature below 32°C -> Fan OFF");
    }
  } else {
    Serial.println("DHT Sensor Error");
  }

  delay(2000);
}
```